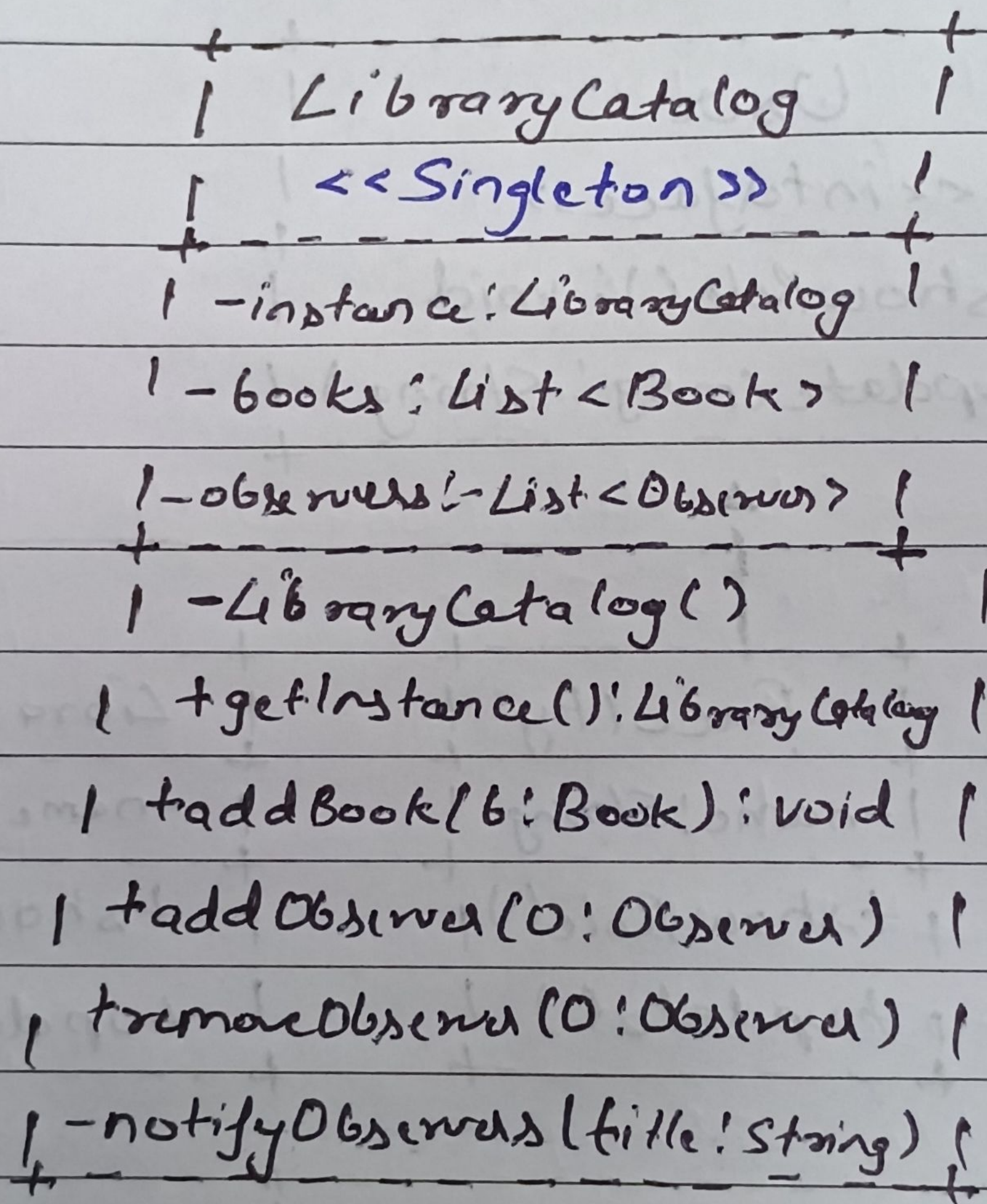


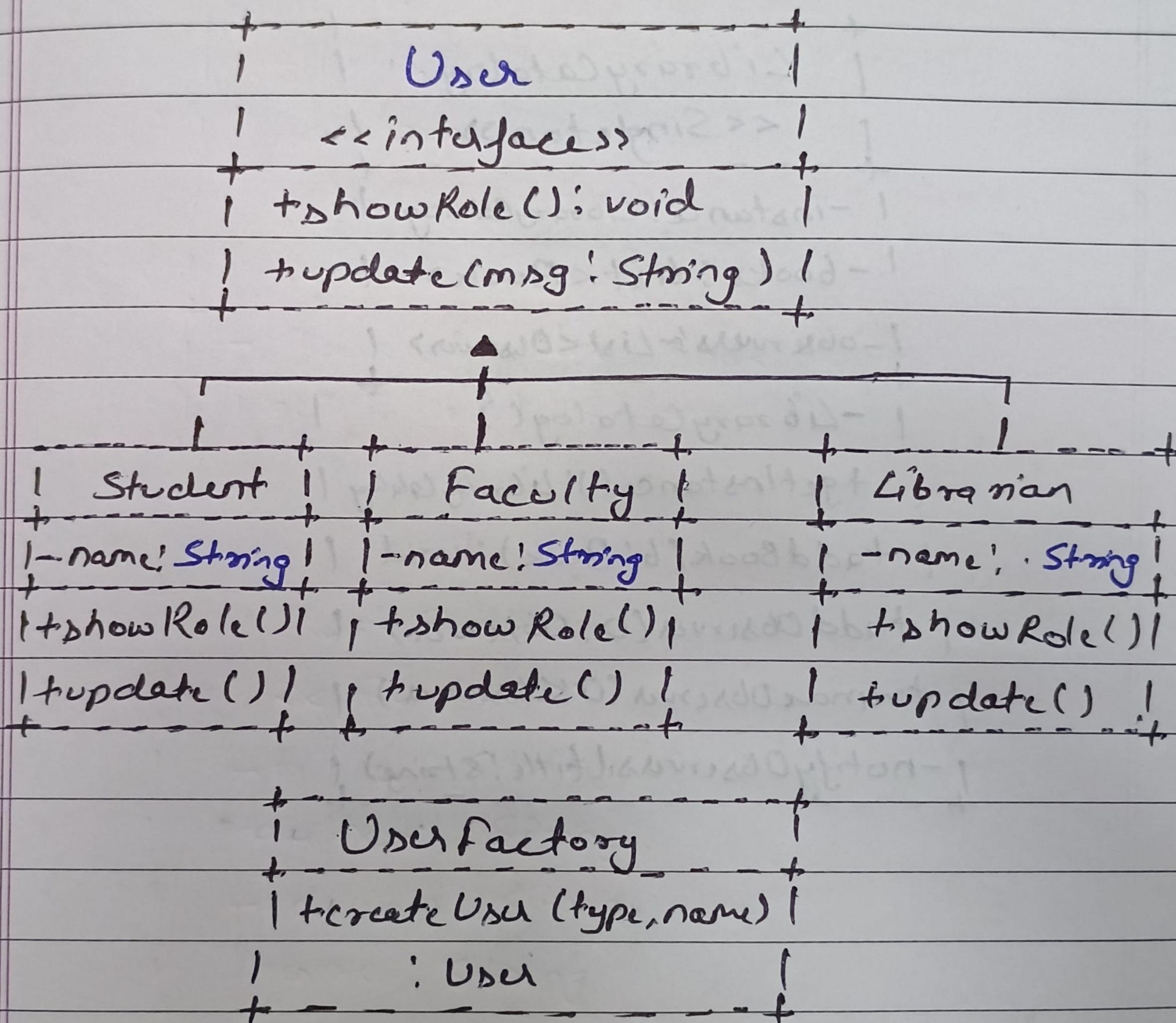
1. Singleton Pattern UML

LibraryCatalog

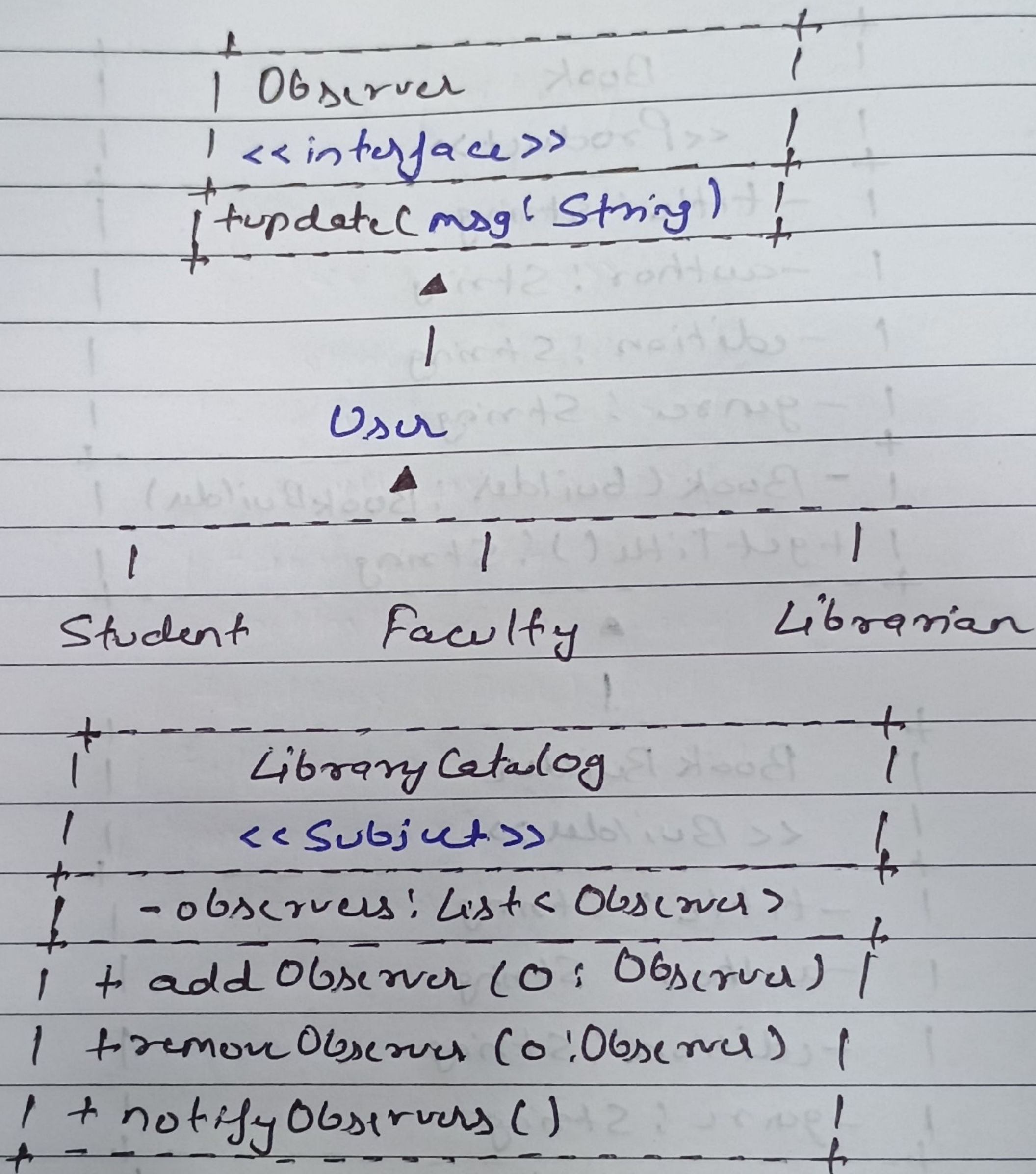


2. Factory Pattern UML

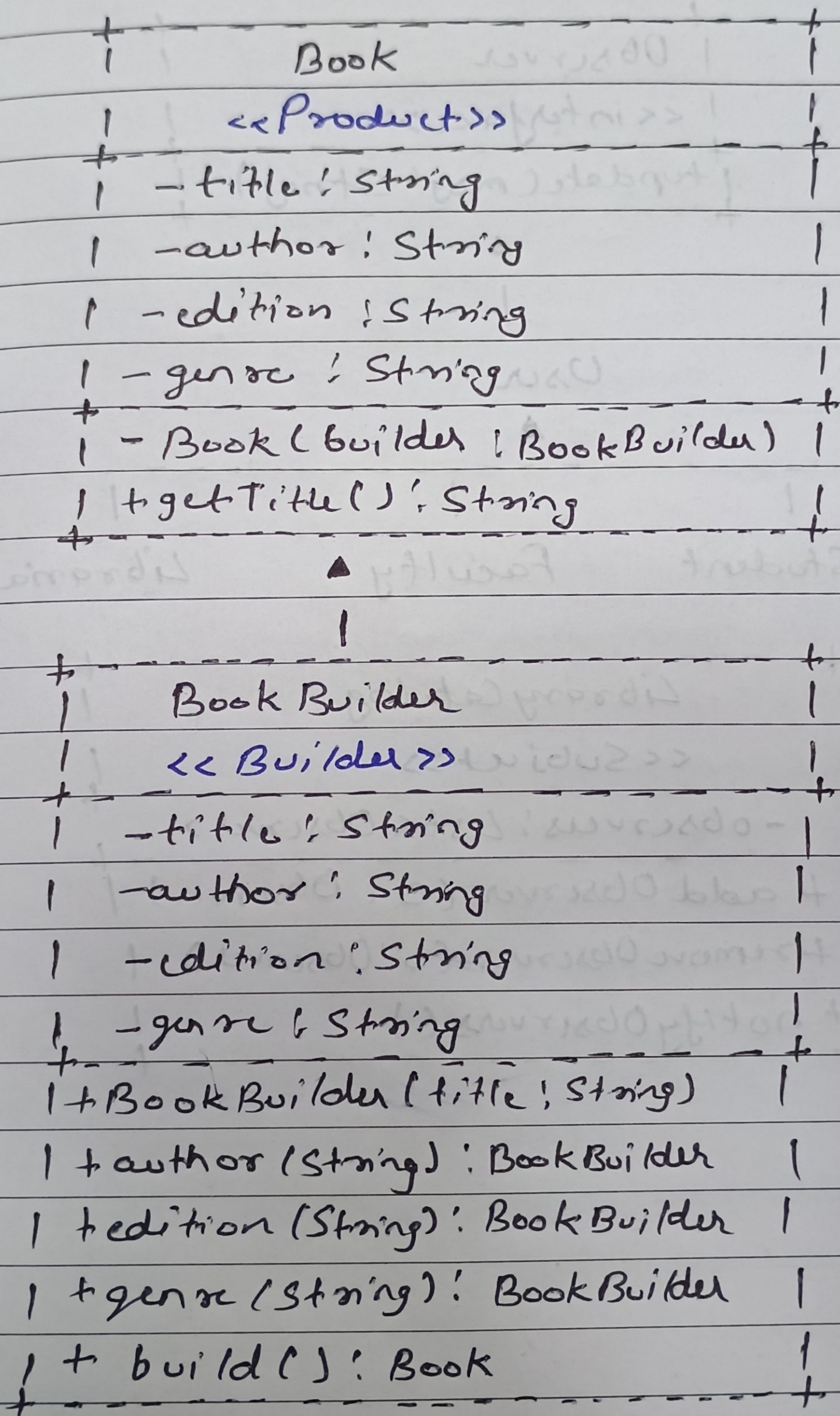
User Creation



3. Observer Pattern UML



4. Builder Pattern UML



5. Integrated UML Diagram

